COMP102 Lecture 7: Graphs I: Representation

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Plan

1. Graphs as Models
2. Direction and Weight
3. Adjacency Lists
4. Adjacency Matrices
5. Representation Tradeoffs
6. Modeling a Graph
7. What Comes Next

Graphs as Models

Graphs are everywhere

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A graph models relationships

A graph is useful when the important information is not only the objects, but **how those objects are connected**.

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Situation	Vertices	Edges
Road map	cities	roads
Social network	people	follows/friendships
Web	pages	links
Code base	files	imports
Courses	courses	prerequisites

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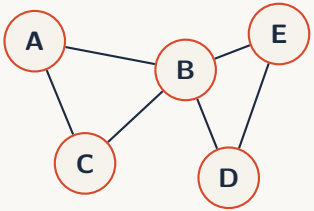

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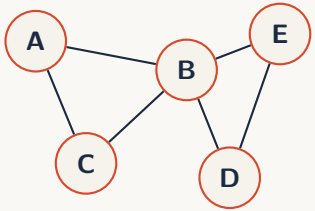

The same drawing can represent roads, friends, webpages, dependencies, or game states. The model decides what nodes and edges mean.

What is a graph?

Definition 1.1 Graph

A graph is a pair $G = (V, E)$ where V is a set of **vertices** and E is a set of **edges** connecting vertices.

What is a graph?

Definition 1.2 Graph

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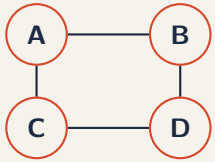
Undirected example



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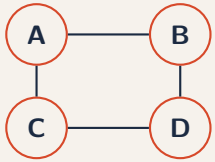
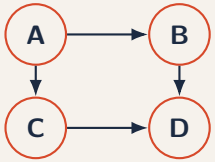


Directed example



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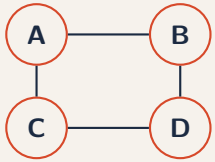

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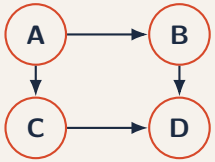
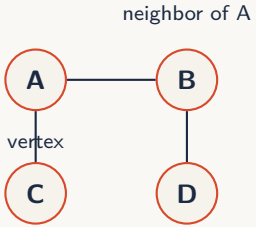


Vocabulary: vertex, edge, neighbor, degree, path, direction, and weight.

Vocabulary I: vertices, edges, neighbors

Words

- A **vertex** is an object.
- An **edge** is a connection.
- A **neighbor** is directly connected to a vertex.

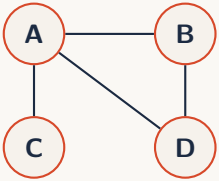


In this graph, the neighbors of A are B and C.

Vocabulary II: degree

Definition 1.5 Degree

The **degree** of a vertex is the number of edges touching it.



Example

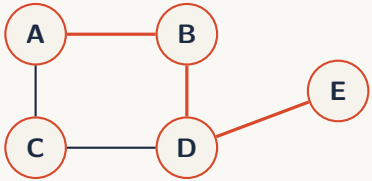
- $\deg(A) = 3$
- $\deg(B) = 2$
- $\deg(C) = 1$
- $\deg(D) = 2$

Degree measures how connected a vertex is.

Vocabulary III: paths

Definition 1.6 Path

A **path** is a sequence of vertices where each consecutive pair is connected by an edge.



Example

$$A \to B \to D \to E$$

is a path.

Length

The length of this path is 3 edges.

Vocabulary IV: direction

Undirected edge

$$A - B$$

means:

A connected to B

and

B connected to A

symmetric relation

Directed edge

$$A \to B$$

means:

A points to B

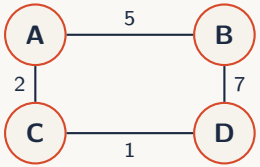
but not necessarily the reverse.

asymmetric relation

Vocabulary V: weights

Definition 1.7 Weight

A **weight** is a value attached to an edge.



Weights can mean

- distance;
- time;
- cost;
- capacity;
- difficulty.

The meaning of a weight comes from the model, not from the graph itself.

Modeling relations

Modeling relations

The hard question

Before choosing code, choose the model:

- What are the vertices?
- What are the edges?
- Are edges directed?
- Do edges need weights?

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A city map can use distance, travel time, toll cost, or safety as the edge weight.

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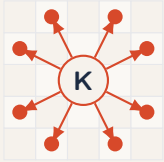
Example: knight moves



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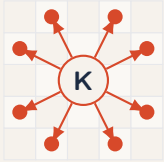

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A city map can use distance, travel time, toll cost, or safety as the edge weight.

Vertices can be objects, positions, configurations, or states. Graph modeling is a translation from a problem to structure.

Direction and Weight

Directed vs undirected

Directed vs undirected

Undirected relation

If A is connected to B, then B is connected to A.

Directed vs undirected

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friendship / two-way road

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Directed relation

If A points to B, B does not necessarily point to A.

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follow / link / prerequisite

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follow / link / prerequisite

For an undirected edge, store both directions. For a directed edge, store only the direction that exists.

Weighted vs unweighted

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Unweighted graph

All edges are treated equally.

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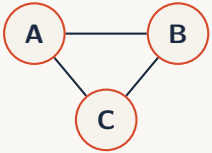

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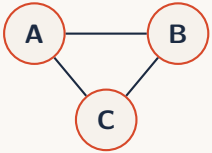

Weighted graph

Each edge carries a value: cost, distance, time, capacity, risk, or probability.

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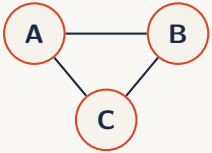

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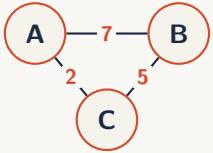

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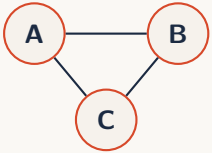

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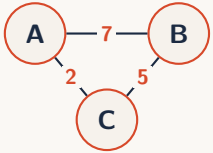
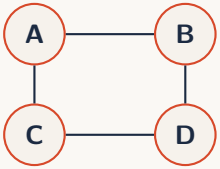


A weight is not automatically a distance. Its meaning comes from the problem model.

Adjacency Lists

Adjacency list



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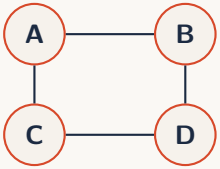
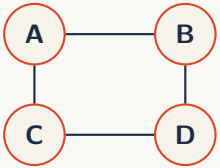


Idea

For every vertex, store the list of its neighbors.

Adjacency list



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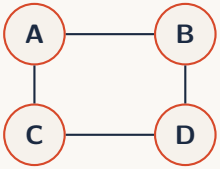
Vertex	Neighbors
A	B, C
B	A, D
C	A, D
D	B, C

Adjacency list



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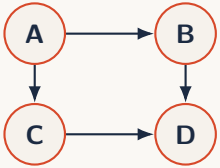
Adjacency lists are the default representation for many graph problems because they store only existing connections.

Directed adjacency lists




```
graph = {
    "A": ["B", "C"],
    "B": ["D"],
    "C": ["D"],
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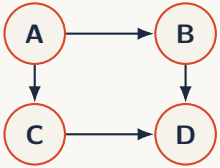
Directed adjacency lists

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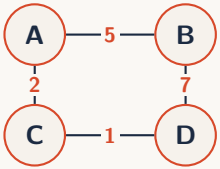
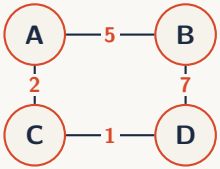
For a directed edge `A -> B`, store `B` in `A`'s list. Do not automatically store `A` in `B`'s list.

Weighted adjacency lists




```python
graph = {
    "A": [("B", 5), ("C", 2)],
    "B": [("A", 5), ("D", 7)],
    "C": [("A", 2), ("D", 1)],
    "D": [("B", 7), ("C", 1)],
}
```

Weighted adjacency lists

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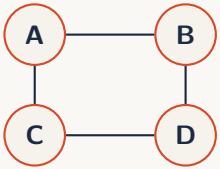
```

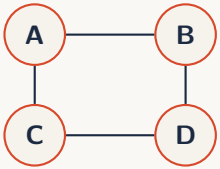
A weighted adjacency list stores extra information with each neighbor: usually `(neighbor, weight)`.

Adjacency Matrices

Adjacency matrix



Adjacency matrix



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Use a 2D table. Entry (i, j) says whether vertex i is connected to vertex j.

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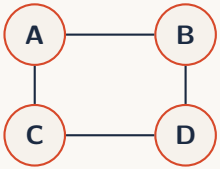

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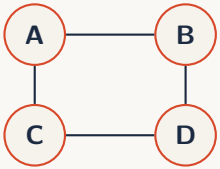
	A	B	C	D
A	0	1	1	0
B	1	0	0	1
C	1	0	0	1
D	0	1	1	0

Adjacency matrix



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	A	B	C	D
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D	0	1	1	0

For an undirected graph, the adjacency matrix is symmetric.

Directed and weighted matrices

Directed and weighted matrices

Directed matrix

Direction breaks symmetry.

Directed and weighted matrices

Directed matrix

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Directed and weighted matrices

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Weighted matrix

Entries store weights, not only 0 or 1.

Directed and weighted matrices

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Weighted matrix

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	A	B	C	D
A	0	5	2	INF
B	5	0	INF	7
C	2	INF	0	1
D	INF	7	1	0

Directed and weighted matrices

Directed matrix

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A	0	5	2	INF
B	5	0	INF	7
C	2	INF	0	1
D	INF	7	1	0

Do not use `0` for “no edge” if zero-weight edges are possible. Use `INF` or `None` depending on the algorithm.

Representation Tradeoffs

List vs matrix

Let $n = |V|$ be the number of vertices and $m = |E|$ the number of edges.

Operation	**Adjacency list**	**Adjacency matrix**
Memory	$O(n + m)$	$O(n^2)$
Check edge $u \to v$	$O(\deg(u))$	$O(1)$
Iterate neighbors of u	$O(\deg(u))$	$O(n)$
Add edge	usually $O(1)$	$O(1)$
Remove edge	depends on structure	$O(1)$

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Adjacency lists are neighbor-friendly and memory-efficient. Matrices are edge-lookup-friendly but pay $O(n^2)$ memory.

Sparse vs dense graphs

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Sparse graph

$m \ll n^2$. Most possible edges are absent.

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Sparse graph

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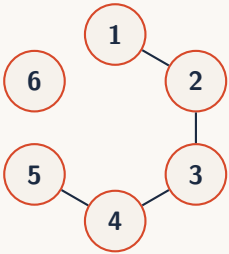

Sparse vs dense graphs

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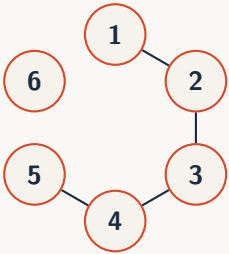

Dense graph

m is close to n^2. Many possible edges exist.

Sparse vs dense graphs

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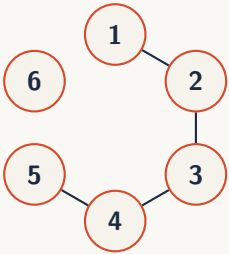

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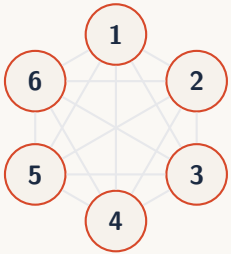

Sparse vs dense graphs

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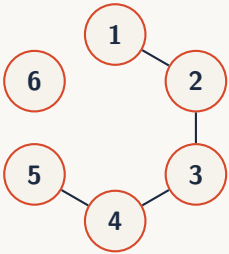

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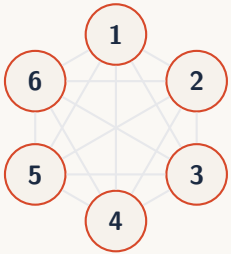

Most real-world graphs are sparse: roads, imports, prerequisites, and many social or web graphs do not contain all possible connections.

Modeling a Graph

Coding: Build a graph

```python
vertices = ["A", "B", "C", "D"]
edges = [
    ("A", "B"),
    ("A", "C"),
    ("B", "D"),
]
```

```python
def build_undirected_graph(vertices, edges):
    graph = {v: [] for v in vertices}

    for u, v in edges:
        graph[u].append(v)
        graph[v].append(u)

    return graph
```

Coding: Build a graph

Goal

Convert a list of input edges into an adjacency list.

```
vertices = ["A", "B", "C", "D"]
edges = [
    ("A", "B"),
    ("A", "C"),
    ("B", "D"),
]
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```
def build_undirected_graph(vertices, edges):
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    graph = {v: [] for v in vertices}

    for u, v in edges:
        graph[u].append(v)
        graph[v].append(u)

    return graph
```

The initialization step is not optional: it preserves isolated vertices.

Coding: Build an adjacency matrix

```
vertices = ["A", "B", "C", "D"]
edges = [
    ("A", "B"),
    ("A", "C"),
    ("B", "D"),
]
```

```
def build_undirected_matrix(vertices, edges):
    n = len(vertices)
    index = {v: i for i, v in enumerate(vertices)}

    matrix = [[0] * n for _ in range(n)]

    for u, v in edges:
        i = index[u]
        j = index[v]
        matrix[i][j] = 1
        matrix[j][i] = 1

    return matrix
```

Coding: Build an adjacency matrix

Goal

Convert a list of input edges into a 0/1 adjacency matrix.

```
vertices = ["A", "B", "C", "D"]
edges = [
    ("A", "B"),
    ("A", "C"),
    ("B", "D"),
]
```

```
def build_undirected_matrix(vertices, edges):
    n = len(vertices)
    index = {v: i for i, v in enumerate(vertices)}

    matrix = [[0] * n for _ in range(n)]

    for u, v in edges:
        i = index[u]
        j = index[v]
        matrix[i][j] = 1
        matrix[j][i] = 1

    return matrix
```

Coding: Build an adjacency matrix

Goal

Convert a list of input edges into a 0/1 adjacency matrix.

```
vertices = ["A", "B", "C", "D"]
edges = [
    ("A", "B"),
    ("A", "C"),
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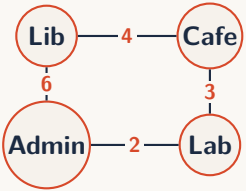
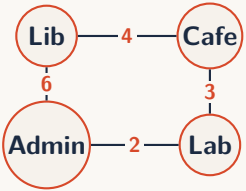
```
def build_undirected_matrix(vertices, edges):
    n = len(vertices)
    index = {v: i for i, v in enumerate(vertices)}

    matrix = [[0] * n for _ in range(n)]

    for u, v in edges:
        i = index[u]
        j = index[v]
        matrix[i][j] = 1
        matrix[j][i] = 1

    return matrix
```

The dictionary `index` converts vertex names into matrix positions.

Case study: campus map



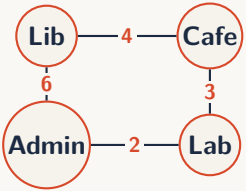
Case study: campus map



Questions

- What are the vertices?
- What are the edges?
- Is the graph directed?
- Should the graph be weighted?
- What could the weights mean?

Case study: campus map



Questions

- What are the vertices?
- What are the edges?
- Is the graph directed?
- Should the graph be weighted?
- What could the weights mean?

Remark 6.1. Representation choice

For a campus map, an adjacency list is usually natural because each place connects to only a few other places.

What Comes Next

What comes next

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Today

We learned how to:

- model relations as graphs;
- distinguish direction and weight;
- store graphs using lists and matrices;
- choose a representation based on tradeoffs.

What comes next

Today

We learned how to:

- model relations as graphs;
- distinguish direction and weight;
- store graphs using lists and matrices;
- choose a representation based on tradeoffs.

Next lecture

Once a graph is represented, we can start asking algorithmic questions:

- Can we reach one vertex from another?
- What order should we explore vertices in?
- What is the shortest path?
- Is the graph connected?